AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 3 · STUDENT LESSON

Classification or Regression?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-2 Week 7: a numeric data type does not determine task meaning.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Identify a task from the meaning of its target. Try the recall before opening notes.

Your learning target

I can identify a task from the meaning of its target.

I can distinguish category agreement from numerical error.

I can specify **output**s and checks that match a user need.

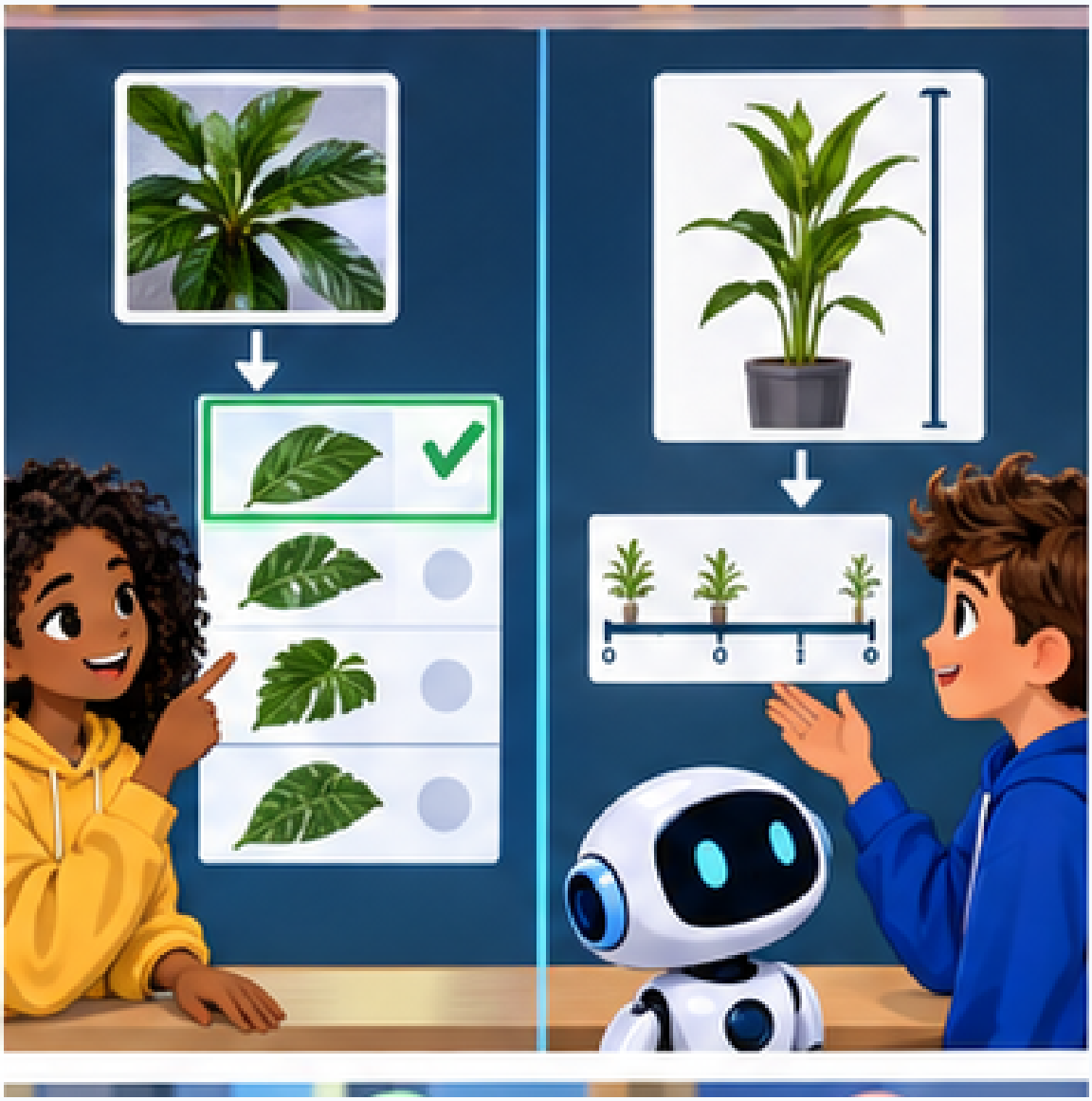
Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Classification predicts a category; **regression** predicts a numerical quantity. The target—not the **input** format—determines the task type.



Your target

LEARNING TARGET System A predicts **class** codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code

0.

Builder name and experiment date:

Readiness for the connected explanation

At dispatch, may the observed arrival duration be used as an **input** to predict that same trip duration? Explain.

This week's end goal

By the end, I can explain **classification** or **regression**? using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

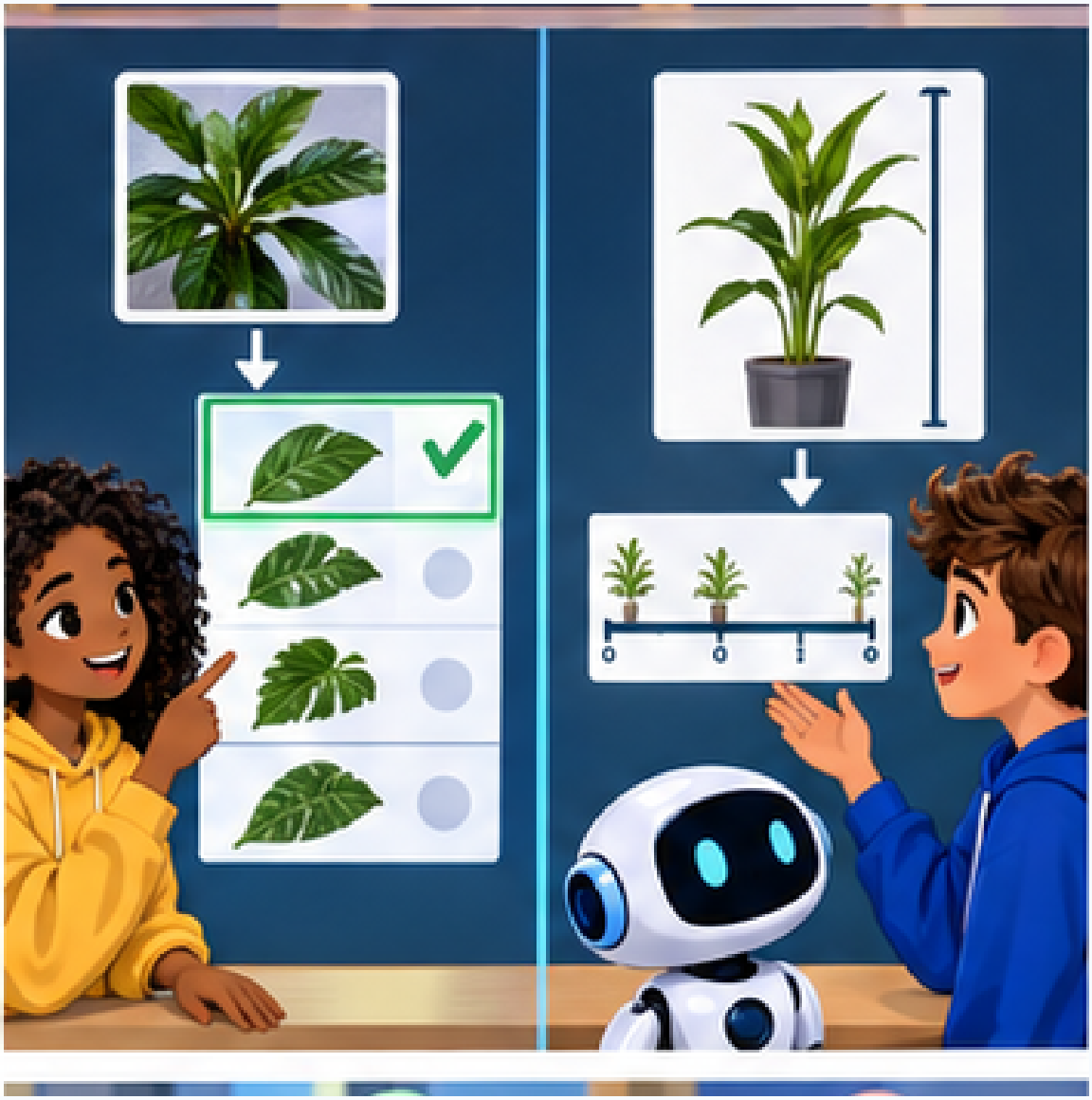
Notice the evidence

> Investigate the supplied Classification or Regression? case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **classification** or **regression**? to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

At robot dispatch, we can ask two questions about the same fictional trip. Will it take more than 10 minutes? Or how many minutes will it take? The first asks for a category. The second asks for an amount with units.

The target question determines the task, even if both models would use the same dispatch-time **input**s.

Question	Target meaning	Task
Will the trip exceed 10 minutes?	0 for no and 1 for yes	Classification
How long will the trip take?	Elapsed duration in minutes	Regression
Which route type will be used?	Indoor or outdoor	Classification

Classification predicts a category from a defined set. **Regression** estimates a numerical quantity for which differences have meaning.

A category may be coded with numbers: 0 and 1 here are names for no and yes, not travel times. Conversely, a duration recorded as a whole number of minutes can still be a **regression** target.

Decimal points do not decide the task.

Keep Week 2's **prediction** time: route length and the saved queue may exist at dispatch, while the observed duration is available later for **training** or evaluation.

Do not feed that observed duration into an earlier dispatch **prediction**.

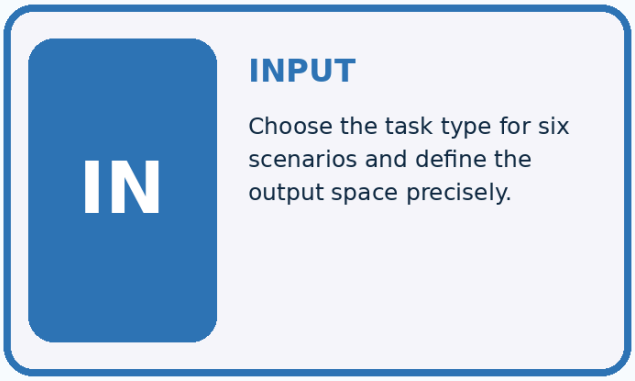
See how it works

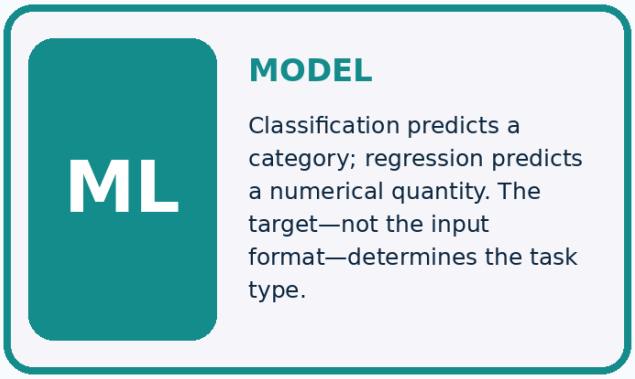
> Follow the mechanism for Classification or Regression?. Connect each labelled input, deciding step and result.

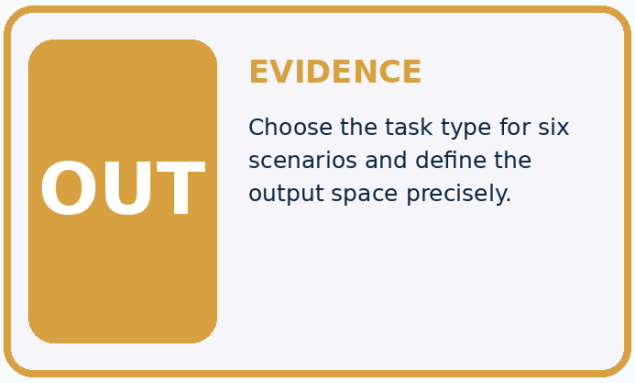
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Choose the task type for six scenarios and define the **output** space precisely.

MODEL IDEA **Classification** predicts a category; **regression** predicts a numerical quantity. The target—not the **input** format—determines the task type.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Predict plant species from a photo: target is a category, so **classification**. Predict plant height next week: target is centimeters, so **regression**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

The following **prediction**s are supplied for practice; no model is fitted here. Each estimate was written before revealing its matching observed duration.

We compare duration error with a category decision made from that estimate. Predict slow when the estimated duration is greater than 10 minutes; exactly 10 is not slow.

A **class**ifier could also predict categories directly, but this example uses a **threshold** on estimates.

```
actual_minutes = [8, 10, 14]
estimated_minutes = [9, 11, 12]
correct = 0
errors = []
for actual, estimate in zip(actual_minutes, estimated_minutes):
    actual_slow = int(actual > 10)
    predicted_slow = int(estimate > 10)
    correct = correct + int(actual_slow == predicted_slow)
    errors.append(abs(estimate - actual))
print(correct, len(actual_minutes))
print(errors)
```

zip pairs the lists in order.

They must have equal lengths and refer to the same trips; otherwise this loop is not a valid evaluation. abs gives the distance from zero, so both a one-minute overestimate and a one-minute underestimate have **absolute error** 1.

The **output** is 2 3 followed by [1, 1, 2]. The mean **absolute error** is (1 + 1 + 2) / 3 = 4/3 minutes, about 1.33 minutes.

Read both kinds of evidence

ACTUAL AND ESTIMATE	DURATION ERROR	CATEGORY MATCH
8 and 9 minutes	1 minute	No and no match
10 and 11 minutes	1 minute	No and yes differ
14 and 12 minutes	2 minutes	Yes and yes match

All three rows use the same greater-than-10 boundary. A small duration error can cross that boundary; a larger error can stay on the same side.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

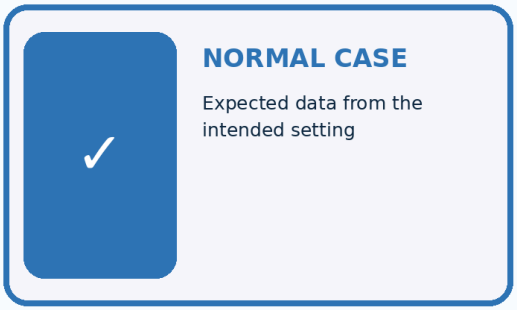
Compare the cases

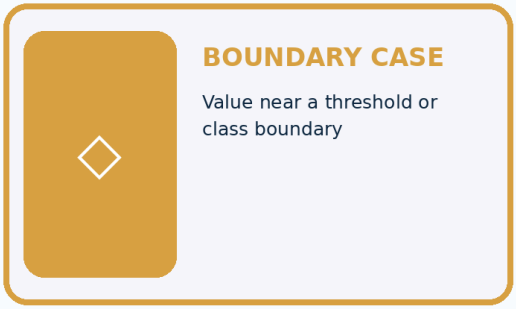
> Compare the supplied cases for Classification or Regression?. Identify the detail that changes the conclusion.

EVALUATION DESIGN

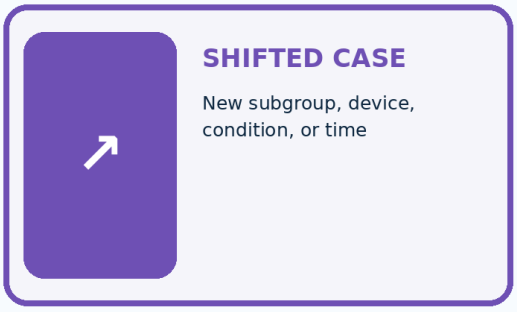
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Change the category boundary to strictly greater than 7 cm.

Predicted result

Prediction: Actual and estimated categories both become short, short, tall: 3/3 correct.

Why the result follows

Reason: The numerical estimates did not improve; the decision definition changed.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Classification or Regression? to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

CLASSIFICATION

predict a category

2

REGRESSION

predict a number

3

CLASS

one allowed category

4

CONTINUOUS TARGET

number that can vary across a range

Term	Precise meaning	My ML example
classification	predict a category	
regression	predict a number	
class	one allowed category	
continuous target	number that can vary across a range	

MISCONCEPTION TO REPAIR A photograph always means **classification** and a table always means **regression**.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Classification	Predicting a category from a defined set
Regression	Estimating a numerical quantity with meaningful differences
Threshold	A boundary used to turn a value into a decision
Absolute error	The nonnegative distance between an estimate and an observation

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Classification or Regression?, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 3 • WORKED EVIDENCE

1. Predict plant species from a photo: target is a category, so classification.
2. Predict plant height next week: target is centimeters, so regression.
3. Predict low/medium/high growth: categories, even though ordered.
4. State allowed classes or numerical unit before selecting a model and metric.

Step 1

STEP 1 Predict plant species from a photo: target is a category, so **classification**.

Step 2

STEP 2 Predict plant height next week: target is centimeters, so **regression**.

Step 3

STEP 3 Predict low/medium/high growth: categories, even though ordered.

Step 4

STEP 4 State allowed **class**es or numerical unit before selecting a model and metric.

CHECK Verify the denominator, unit, split, **class** definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Worked explanation · Classification or Regression?

The 14-minute trip has the largest duration error in our example, yet its slow category is correct. The 10-minute trip has only a one-minute error but the category is wrong. These checks answer different questions.

Report 2 correct out of 3 category decisions separately from a mean **absolute error** of about 1.33 minutes. Do not compare a fraction directly with minutes.

A trip actually takes 10 minutes and its estimate is 10.5 minutes. At the greater-than-10 boundary, give the actual category, predicted category and absolute duration error. Is the category decision correct?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Classification or Regression?

Distinguish predicting growth height from predicting a growth category.

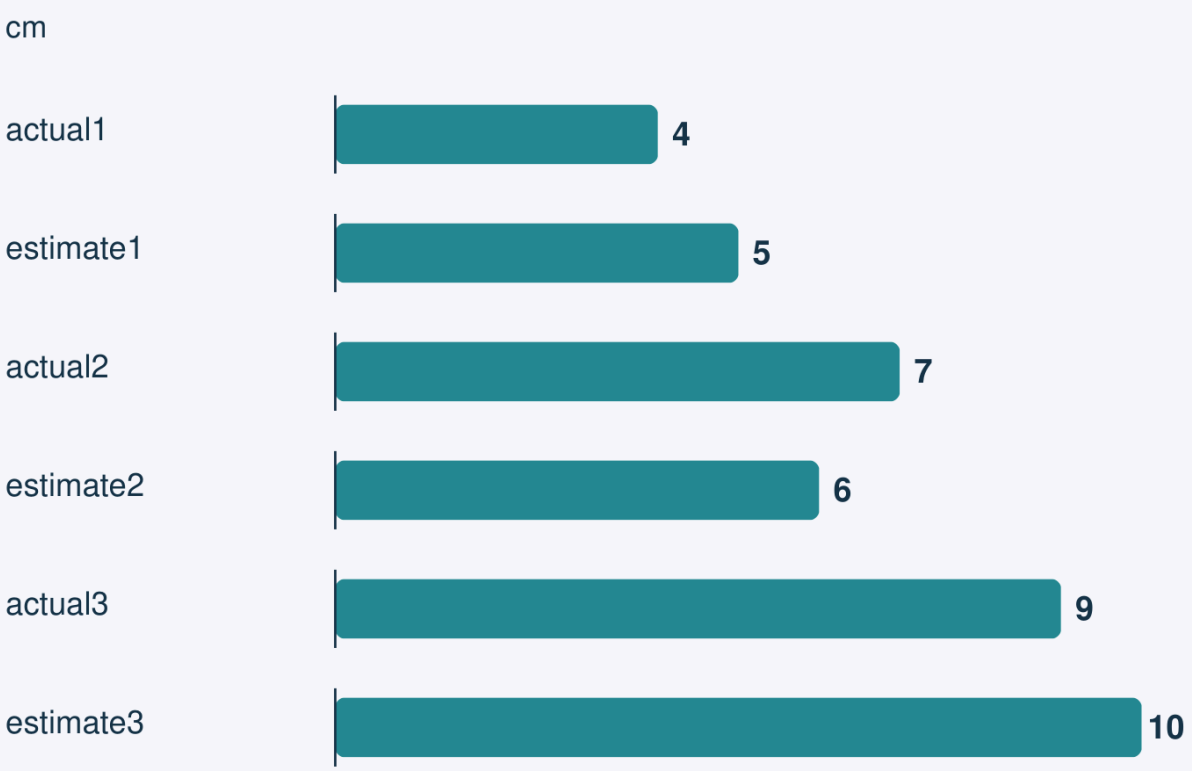
The facts we will use

> **Supplied facts**
>
> Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.

AI ACADEMY / AI-4/3

Actual and estimated heights (paired order)



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/3

Classification or Regression?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Regression errors

Absolute errors are 1, 1, 1 cm; mean absolute error is 1 cm.



2. Actual categories

Actual classes are short, tall, tall.



3. Predicted categories

Estimated classes are short, short, tall: 2/3 correct.

CHANGE ONE THING

Change the category boundary to strictly greater than 7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Regression errors	Absolute errors are 1, 1, 1 cm; mean absolute error is 1 cm.
2. Actual categories	Actual classes are short, tall, tall.
3. Predicted categories	Estimated classes are short, short, tall: 2/3 correct.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Regression error and category accuracy measure different targets; neither alone covers every use consequence.

Keep the stated limit

Read the labelled picture: Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Change the category boundary to strictly greater than 7 cm. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Classification or Regression? claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A photograph always means classification and a table always means regression. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A photograph always means **classification** and a table always means **regression**.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

A task specification should name the **prediction** moment, permitted **input**s, target meaning and how **prediction**s will be checked. **Classification** checks compare category decisions with observed categories.

Regression checks measure numerical error with stated units. Later weeks examine additional metrics and why the consequences of different errors matter.

Check the claim against the facts

A correct category does not imply an exact quantity, and low numerical error does not guarantee every **threshold** decision is right. Three supplied examples establish only these calculations.

They do not establish a trained model's reliability or show that a robot should act without review.

A team needs an estimated arrival duration in minutes. State the task type and a suitable error check. Explain why reporting only the fraction of correct slow or not-slow decisions leaves its main question unanswered.

STUDENT LESSON · 8 OF 18

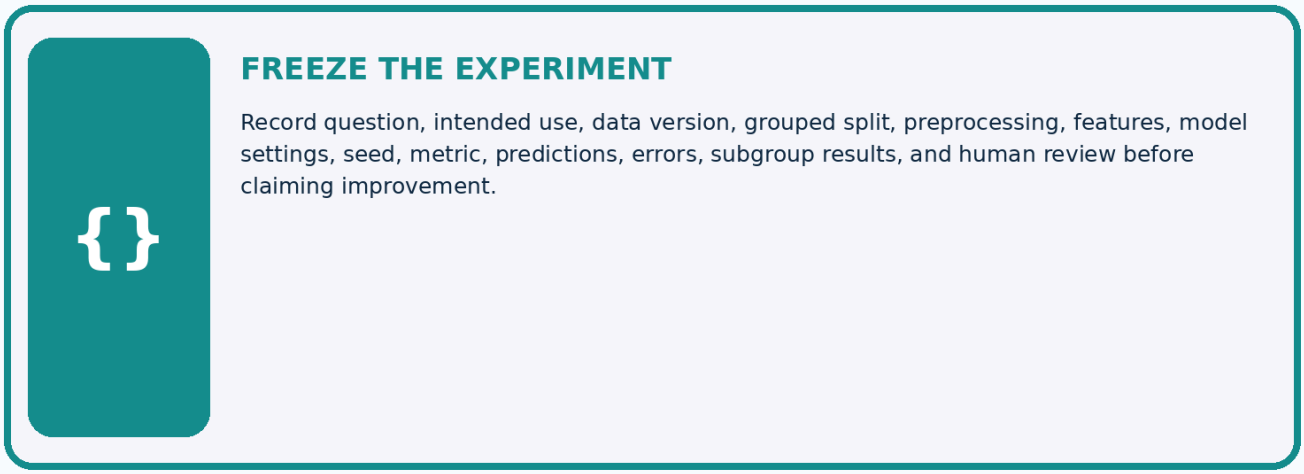
Practise together

Practise Classification or Regression? with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success **threshold**

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive **class** + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

A learner says that predicting route type is **regression** because indoor is encoded as 1 and outdoor as 2. The numbers are category codes: their difference does not measure travel time or another meaningful amount.

Treat this as **classification** and judge the predicted category. Changing the codes must not change the meaning of the task.

Another learner puts durations into short and long bins, then claims to predict exact minutes. Binning creates a **classification** target and loses within-bin detail.

If the user needs minutes, keep a quantity target and evaluate error in minutes. If the user needs a **threshold** decision, define the boundary and relevant consequences. Neither task is automatically better.

Practise with less help: Trace both checks

Use workbook W2 for the connected example "**Classification** and **regression**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Classification** and **regression**. Proposed explanation: "Numbers always mean **regression**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

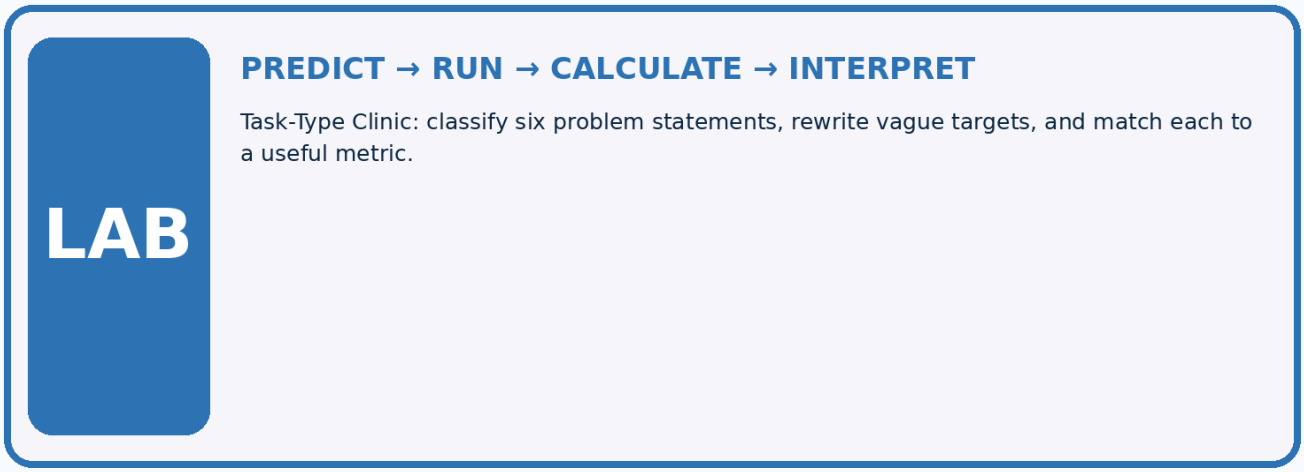
Run the investigation

> Investigate Classification or Regression?. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Task-Type Clinic: **class**ify six problem statements, rewrite vague targets, and match each

to a useful metric.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Boundary
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Missing
Prediction: _____
Actual: _____
Calculation / evidence: _____

> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

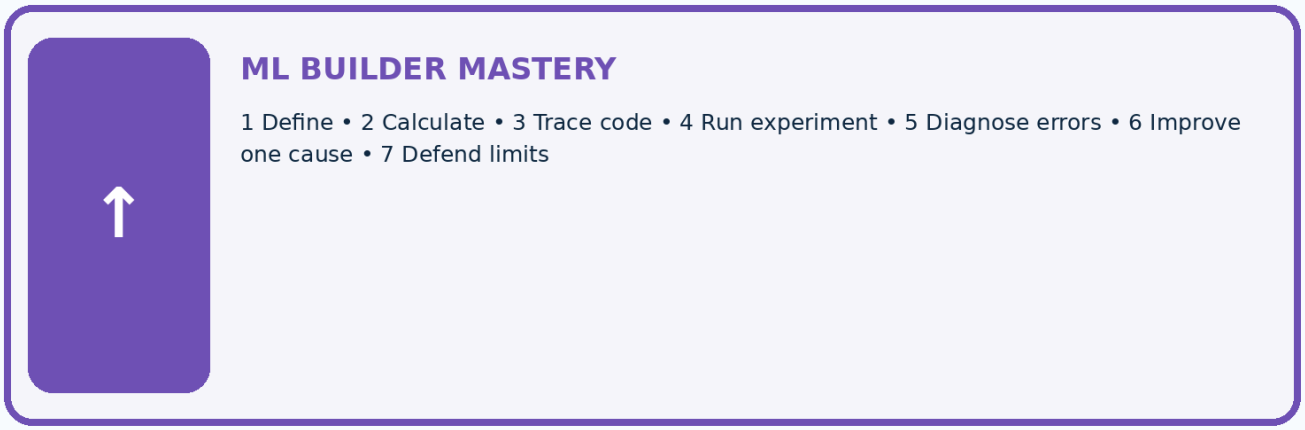
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Classification or Regression?. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **classification**, **regression**, **class**, **continuous target**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A photograph always means **classification** and a table always means **regression**.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is

completed independently.

STUDENT LESSON · 11 OF 18

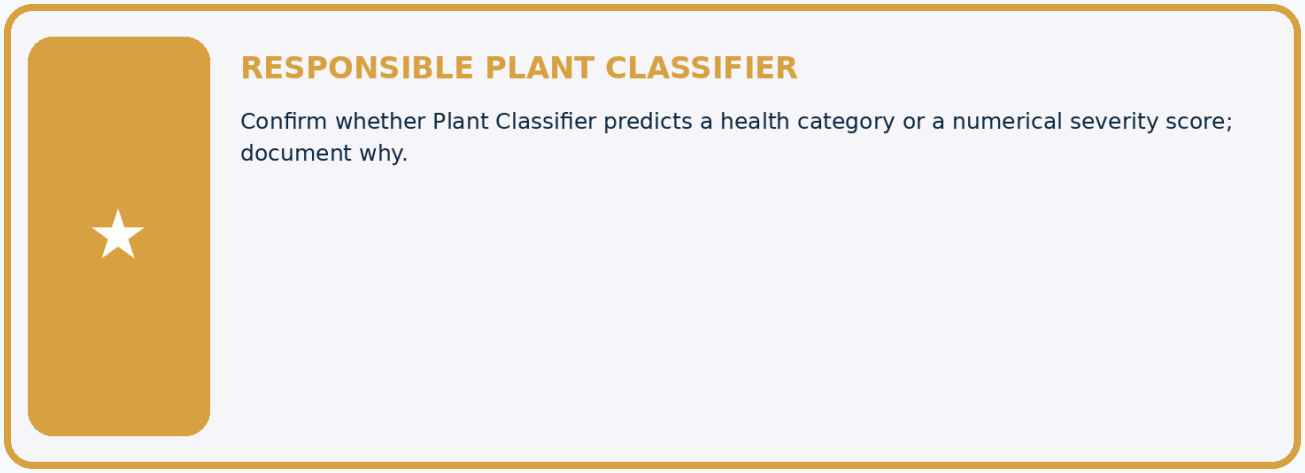
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Confirm whether Plant **Class**ifier predicts a health category or a numerical severity score; document why.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this **class**room prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

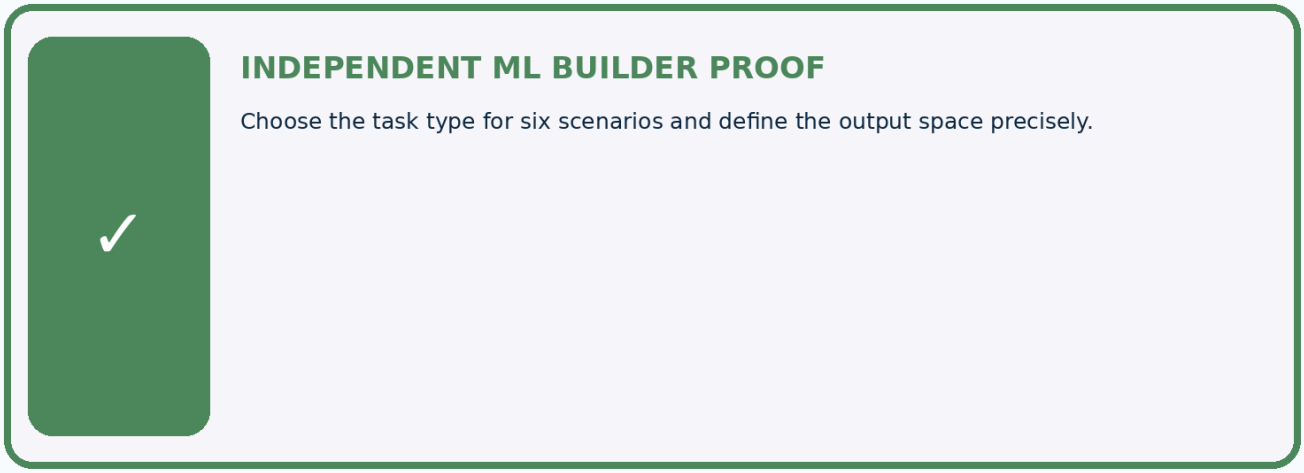
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Choose the task type for six scenarios and define the **output** space precisely.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.

> ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Python and Table Readiness.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Classification or Regression? case for supported practice. Repeated examples do not become fresh assessments.

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Task A predicts healthy/spotted; B predicts next week's height in cm.

The question

Choose task and **output** types.

Worked explanation

A is **classification** with categorical **output**; B is **regression** with numeric **output** and units. Numeric **class** codes do not make **classification regression**.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Choose the task by the meaning of the output

Evidence or stage	Value or result
Healthy or spotted	Classification, category
Height next week	Regression, numeric cm
Class code 0 or 1	Still a category

Task A predicts healthy/spotted; B predicts next week's height in cm.

Choose task and **output** types.

A is **classification** with categorical **output**; B is **regression** with numeric **output** and units. Numeric **class** codes do not make **classification regression**.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Compare new evidence independently

For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes.

Write actual and predicted category lists, count correct categories, calculate each **absolute error** and the mean error in minutes. State what each result measures.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Task A predicts whether a plant is fern or grass. Task B predicts its height in centimetres. Which is **classification** and which is **regression**?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Confirm whether Plant **Class**ifier predicts a health category or a numerical severity score; document why.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

System A predicts **class** codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Classification or Regression?, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Distinguish predicting growth height from predicting a growth category.

By the end, I can	How I will show it
Identify a task from the meaning of its target	Explain the deciding step, show a trace or test, and state one limit.
Distinguish category agreement from numerical error	Explain the deciding step, show a trace or test, and state one limit.
Specify outputs and checks that match a user need	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several **class**es.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Classification or Regression? without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Identify a task from the meaning of its target

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Distinguish predicting growth height from predicting a growth category.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Estimated **class**es are short, short, tall: 2/3 correct.

1. Actual categories

2. **Regression** errors

3. Predicted categories

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Predicted categories Estimated **class**es are short, short, tall: 2/3 correct. Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the category boundary to strictly greater than 7 cm. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Actual and estimated categories both become short, short, tall: 3/3 correct.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Actual and estimated categories both become short, short, tall: 3/3 correct. The numerical estimates did not improve; the decision definition changed.

Repair a missing building block

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

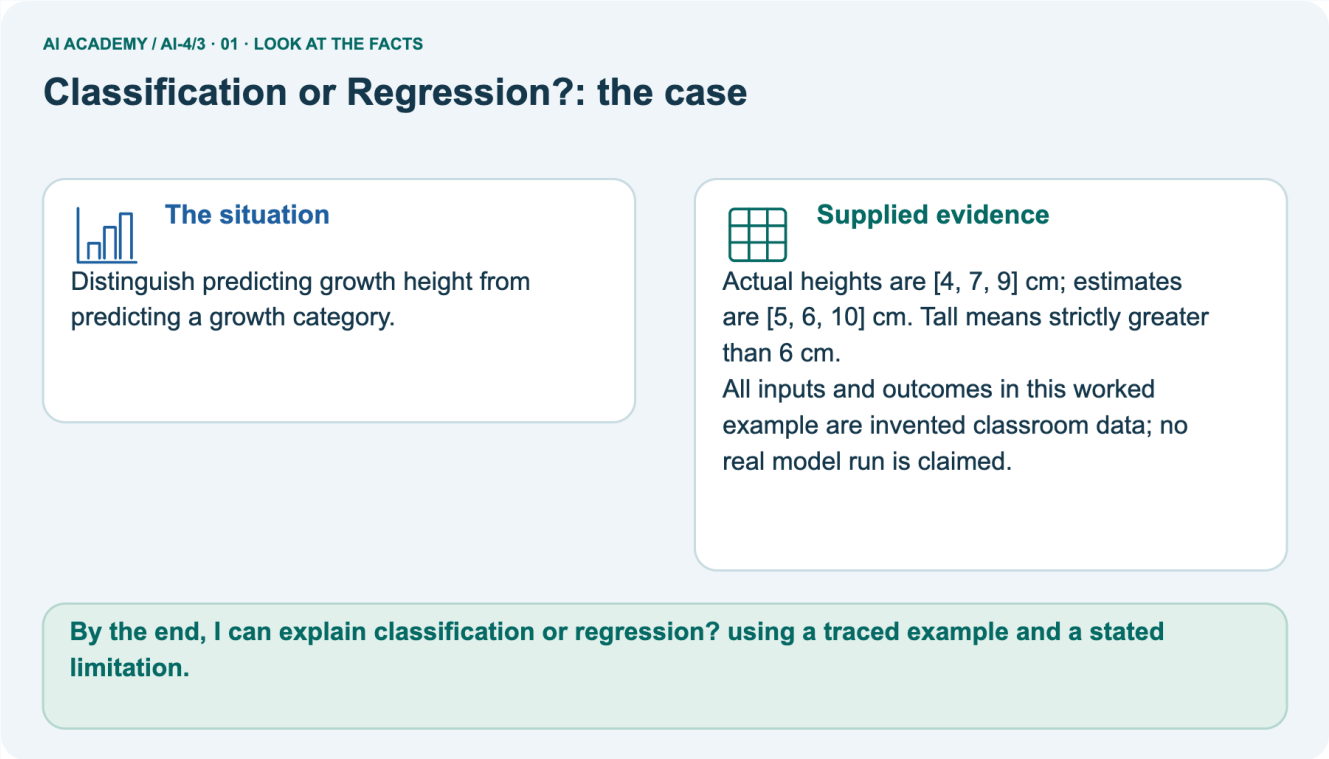
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Classification or Regression? graphic using its labels, deciding evidence and relevant limitation.

01. Classification or Regression?: the case



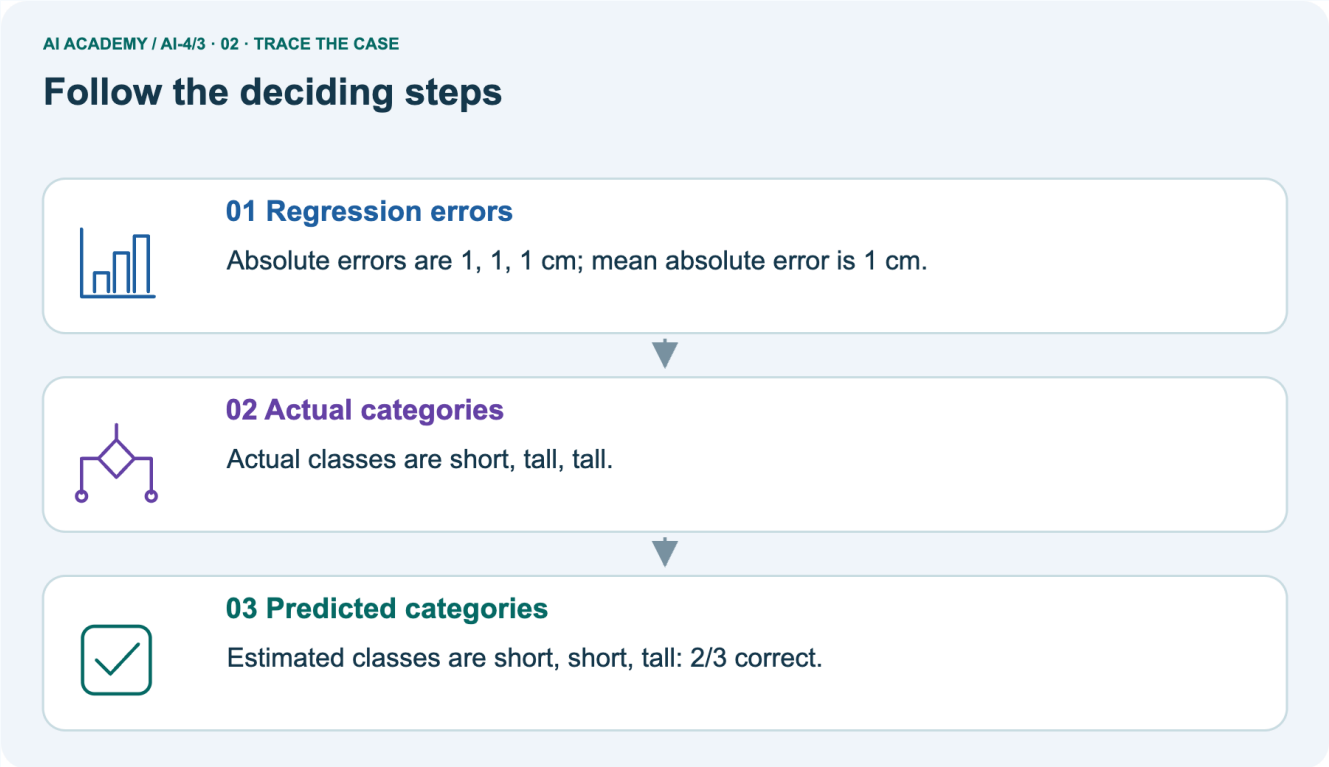
Follow the labelled picture with the matching explanation. Open the vector diagram

Distinguish predicting growth height from predicting a growth category. Supplied facts: Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Regression errors: **Absolute error**s are 1, 1, 1 cm; mean **absolute error** is 1 cm. → Actual categories: Actual **class**es are short, tall, tall. → Predicted categories: Estimated **class**es are short, short, tall: 2/3 correct.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Actual and estimated heights (paired order)

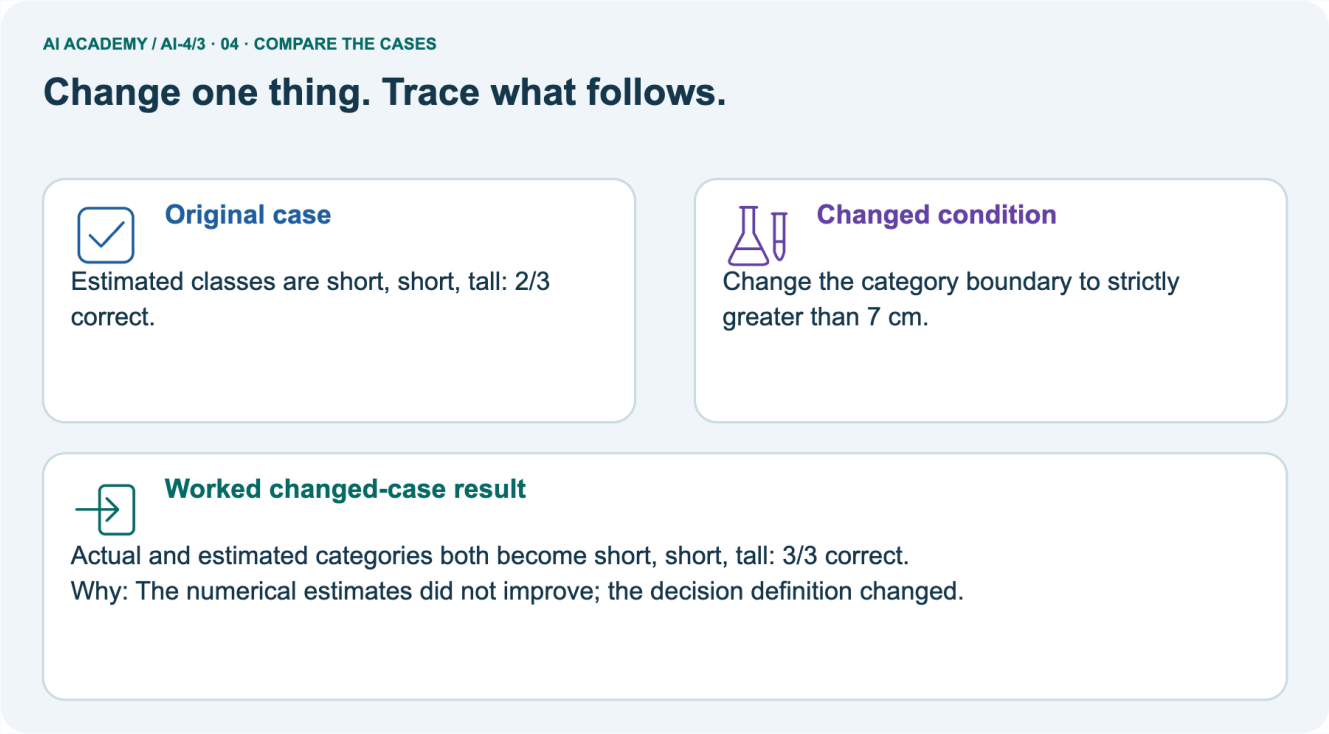


Follow the labelled picture with the matching explanation. Open the vector diagram

Actual and estimated heights (paired order). Units: cm. actual1: 4. estimate1: 5. actual2: 7. estimate2: 6. actual3: 9. estimate3: 10.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



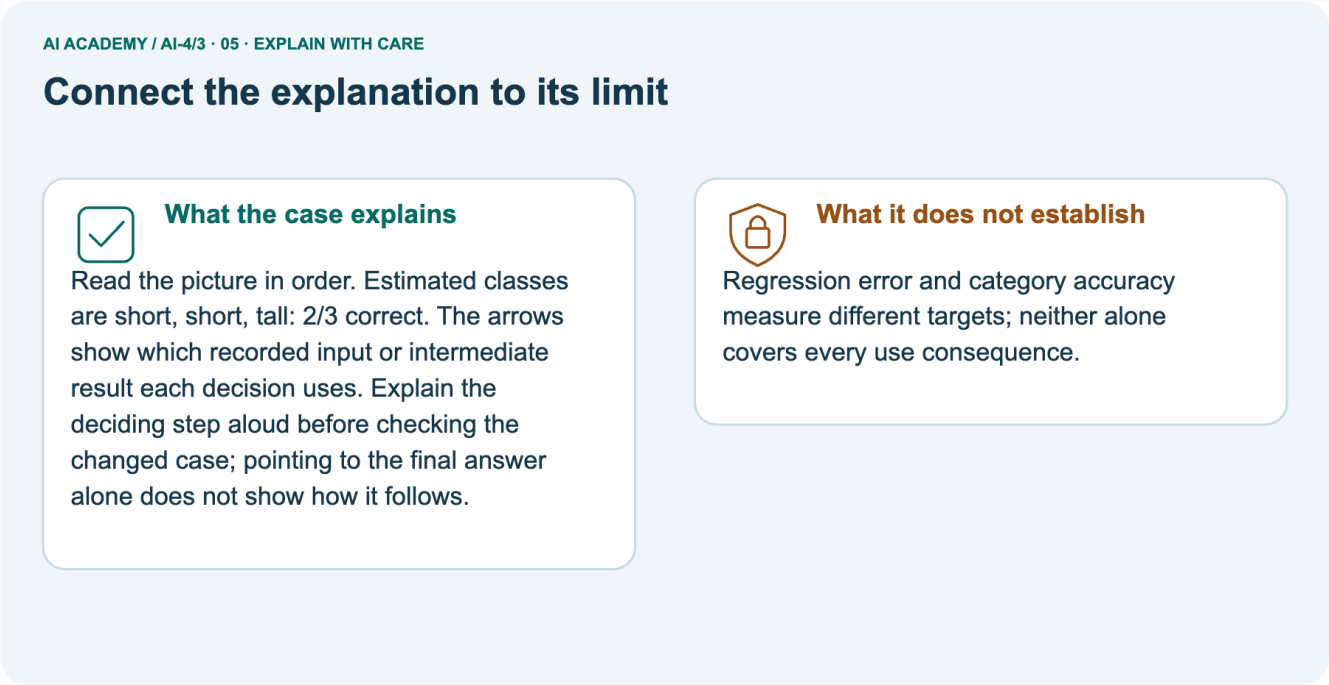
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Estimated **class**es are short, short, tall: 2/3 correct. Changed condition: Change the category boundary to strictly greater than 7 cm.

Expected result: Actual and estimated categories both become short, short, tall: 3/3 correct. Reason: The numerical estimates did not improve; the decision definition changed.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

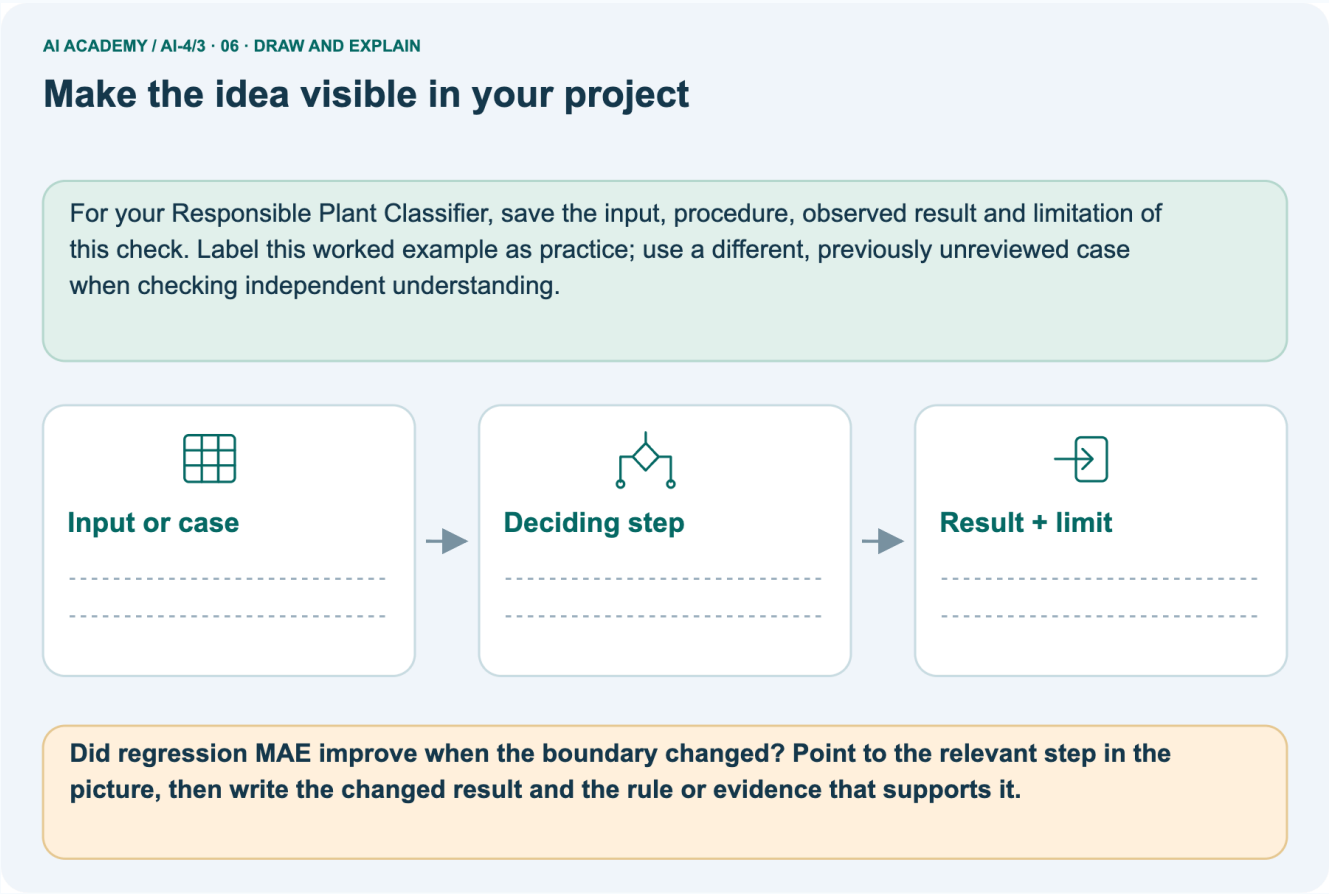
Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Regression** error and category accuracy measure different targets; neither alone covers every use consequence.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.